

Pushkar Shinde

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TECHNICAL SKILLS

Languages: Java, C, SQL.

Frameworks: Spring Boot, Spring Cloud Gateway, Spring Cloud Config, Eureka.

Tools & Technologies: Docker, Podman, Keycloak, RabbitMQ, Maven, Git, PostgreSQL.

PROJECTS

AI-Powered Fitness Application | [GitHub Link](#)

June 2025 - Feb 2026

- **Tech Stack:** Java, Spring Boot, Microservices, Podman, Keycloak, PostgreSQL.
- Architected a scalable multi-microservices fitness platform with carefully designed stateless authentication flow with **JWT and Keycloak** to support horizontal scalability.
- Integrated Gemini API to generate personalized workout plans using prompt-engineered **NLP pipelines**; validated response consistency through structured test cases.
- Implemented asynchronous communication using **RabbitMQ** to decouple services and reduce request latency during concurrent traffic by around 50%.
- Configured **Spring Cloud Config** for centralized configuration and an API gateway as a single point of contact.
- Resolved memory exhaustion problem by: migrated from Docker to **Podman**; applied per-container memory limits; enabled **Spring Boot lazy initialization**; turned on **OS VM**. Reduced peak RAM usage by nearly 30%.

Cause List Converter | [GitHub Link](#)

Feb 2026 – Apr 2026

- **Tech Stack:** Java, Spring Boot, TypeScript, PostgreSQL, Apache POI, Podman.
- Engineered an AI-powered full-stack pipeline that converts unstructured **Allahabad High Court cause list** PDFs into formatted multi-sheet Excel workbooks; extracts 12 structured fields across dynamically detected legal section types.
- Designed a chunking algorithm using PDFBox that splits PDFs at legal headers, enabling reliable structured extraction via Gemini 2.5 Flash with a verbatim-engineered system prompt, achieving 100% parsing accuracy.
- Resolved Gemini free-tier rate limiting by implementing a **thread-safe round-robin API key pool** across N keys, reducing effective inter-chunk delay from 13s to 2s and cutting average processing time by 40%.
- Built **exponential backoff retry** logic eliminating 80% data drops on unstable API days.

Expense Tracker | [GitHub Link](#)

Nov 2025 - Nov 2025

- **Tech Stack:** C, AES Encryption, OpenSSL (EVP API).
- Engineered a secure, CLI finance manager supporting CRUD operations, substring search, and monthly records.
- Architected a zero-plaintext binary persistence layer utilizing **AES-256-CBC** combined with PBKDF2 (100,000 iterations) and dynamic 16-byte salting to prevent brute-force and rainbow-table attacks.
- Resolved critical CLI buffer overflow crashes and plaintext memory leaks by migrating from `scanf` to strictly bounds-checked `fgets`, engineering custom float validation logic (`strtod`). Eliminated 100% of malicious input crashes and reduced unprotected memory footprint to 0 bytes.

EDUCATION

Motilal Nehru National Institute of Technology Allahabad, Prayagraj

Aug 2024 - Present

Master of Computer Applications | CPI: 8.15

Institute of Science, Nagpur

Sept 2020 - June 2023

Bachelor of Science | PCM | Percentage: 67.1

POSITIONS OF RESPONSIBILITY

Application Club MNNIT | Lead

Oct 2025 – Present

- Guided junior students through their academic skill-building journey. Executed technical events, including organizing **The Daemon**, an interactive **Linux treasure hunt game** designed to gamify command-line education.

National Cadet Corps | Sergeant

June 2020 – Aug 2023

- Mentored Junior Division cadets on firing principles and conducted map-reading demonstrations during CATC (2021) at unit 3 Maharashtra Engineers Company NCC Nagpur.
- Led a 10-member team at the BEG Pune army attachment camp (2022), earning "The Most Active Team" award for exemplary performance.